

AI-POWERED LOAN DEFAULT PREDICTION SYSTEM

Explainable machine-learning risk scoring that helps banks predict borrower defaults early.

COVER

- **Project Title:** AI-Powered Loan Default Prediction System
- **Team ID:** 36
- **Institute / Course:** GLA University, Mathura — B.Tech CSE (AIML and IOT), Major Project (Semester 7)
- **Version:** v1.0
- **Date:** 04 Oct 2026
- **Github Repo:** <https://github.com/kkkkushagra/AI-Powered-Loan-Default-Prediction-System>

Revision History

Version	Date	Author	Change
v0.1	01 Oct 2026	Kushagra Trivedi	Initial Draft
v1.0	04 Oct 2026	Kushagra Trivedi	Complete Synopsis Submitted

1. Overview

- **Problem Statement:** Banks and NBFCs lose a large share of their profit to loan defaults, yet most credit decisions still rely on static scorecards and manual judgement that react late and miss non-linear risk patterns. Indian bank asset quality is currently strong (the RBI puts the gross NPA ratio of scheduled commercial banks at roughly 1.8% at March 2026), but its stress tests show the ratio could rise to about 3.2%–4.2% under adverse scenarios [1][2]. Lenders need a way to spot risky borrowers before disbursal and during the loan lifecycle, and to explain why.
- **Proposed Solution:** An AI-powered banking tool that ingests applicant data, engineers credit-risk features, and scores each borrower's probability of default in real time using a gradient-boosted ML model. Every score is returned with a risk band (Low / Medium / High) and the top factors behind it (SHAP), and is surfaced on a loan-officer dashboard and through a REST API that can plug into existing loan-origination systems.
- **Goal:** Reduce credit losses by helping lenders identify high-risk borrowers early, automate risk scoring, and make transparent, data-driven lending decisions.

- **Non-goals:** Actual loan disbursement or payments, live credit-bureau integration, fully automated approve/decline decisions (a human stays in the loop), and production regulatory certification in v1.
- **Value Proposition:** Faster and more consistent risk assessment than manual review, better default detection than a static scorecard, and explanations that loan officers and auditors can actually read.
- **Real-world application:** A bank reduces loan defaults by about 15% by using AI-based risk scores to tighten approvals, price risk, and prioritise early collections for flagged accounts. This is an illustrative scenario; our project will measure its own uplift by back-testing on held-out data.

Key features at a glance

- ML risk prediction model
- Data preprocessing and feature engineering pipeline
- Real-time scoring
- Loan officer dashboard
- Historical trend analysis
- API integration with loan systems
- Explainable predictions (SHAP)

2. Scope and Control

➤ 2.1 In-scope

- Data ingestion (CSV upload and REST), validation, cleaning and feature engineering pipeline.
- Model training, comparison, calibration and versioning (Logistic Regression baseline, Random Forest, XGBoost, LightGBM).
- Real-time scoring API with risk bands and SHAP-based reason codes.
- Role-based loan officer dashboard: applicant lookup, risk score, explanations, portfolio and historical trend views.
- Model performance and data-drift monitoring; outcome feedback for retraining.

➤ 2.2 Out-of-scope

- Real bank data, live credit-bureau (CIBIL/Experian) integration, loan disbursement and repayment processing, mobile apps, and regulatory certification in v1.

➤ 2.3 Assumptions

- Public credit datasets are sufficiently representative to demonstrate the approach; results are indicative, not production-grade.

- Loan officers use the score as decision support, not as an automatic decision.

➤ 2.4 Constraints

- About 10 weeks in Semester VII, free-tier cloud hosting, and public/synthetic data only.

➤ 2.5 Dependencies

- Kaggle/UCI datasets, Python ML libraries (scikit-learn, XGBoost, LightGBM, SHAP), PostgreSQL, a container host (Render/AWS free tier), and GitHub Actions for CI/CD.

➤ 2.6 Acceptance criteria and sign-off

- GIVEN a valid applicant record WHEN it is posted to /api/v1/score THEN a default probability, risk band and top-5 reasons are returned in ≤ 300 ms (p95).
- GIVEN a loan officer WHEN they open an applicant on the dashboard THEN the score, risk band and SHAP explanation chart are visible within 2 seconds.
- GIVEN the held-out test set WHEN the final model is evaluated THEN AUC-ROC ≥ 0.78 and recall on defaulters $\geq 70\%$ at the chosen threshold.
- **Sign-off:** Mentor approves the demo and test report; all P1 bugs closed.

Sign-off Table

Stakeholder	Role	Decision Area	Signature/Approval	Date
Mentor Name	Mentor	Scope, Final Acceptance		
Kushagra Trivedi	Project Lead	Release readiness		

3. Stakeholders and RACI

Activity	Responsible	Accountable	Consulted
Requirements, Data Collection, Preprocessing and Feature Engineering	Kushagra Trivedi	Kushagra Trivedi	Mentor
Model Development, Explainability, Backend API and Database	Kushagra Trivedi	Kushagra Trivedi	Mentor
Dashboard (Frontend), Testing and Documentation	Kushagra Trivedi	Kushagra Trivedi	Mentor
Deployment and Release	Kushagra Trivedi	Kushagra Trivedi	Mentor

4. Week-wise Plan and Assignments

Week	Dates	Milestone	Tasks	Deliverable	Status
1	04-10 Oct	Requirements & Data Understanding	Study problem statement; define scope and objectives; collect and understand dataset; literature review and technology selection	SRS Draft	Planned
2	11-17 Oct	Data Pipeline	Data collection and preprocessing; data cleaning and imputation; exploratory data analysis (EDA)	Preprocessing Module + EDA Report	Planned
3	18-24 Oct	Feature Engineering	Feature analysis and selection; create engineered features (Ratios, WoE, IV); prepare training and testing datasets	Feature Pipeline	Planned
4	25-31 Oct	Baseline Models	Implement baseline ML models (Logistic Regression, Random Forest); evaluate and compare performance; create model comparison report	Baseline Report	Planned

5	01-07 Nov	Advanced Models	Implement advanced models (XGBoost/LightGBM); hyperparameter tuning; select best-performing model	Model v1 (MLflow)	Planned
6	08-14 Nov	Explainability & Scoring	Implement explainability (SHAP, feature importance); build risk-scoring logic; analyze results and insights	Explainable Scoring Report	Planned
7	15-21 Nov	Dashboard & Trends	Design and develop dashboard UI; integrate model predictions; add charts and analytics	Dashboard Demo	Planned
8	22-28 Nov	Integration & Monitoring	Integrate backend and frontend; implement user authentication if required; add monitoring and logging	Integrated Build	Planned
9	29 Nov-05 Dec	Testing & Hardening	Perform end-to-end testing; fix bugs and improve performance; validate scoring and reports	Test Report	Planned
10	06-12 Dec	Release & Final Demo	Deploy application; prepare project report and presentation slides; final presentation and demo	v1.0 + Final Report + Presentation Deck	Planned

5. Users and UX

➤ 5.1 Personas

- **Loan Officer:** Reviews 30–50 applications a day; wants a clear risk score and plain-language reasons so she can decide quickly and justify it.
- **Risk Manager:** Monitors portfolio health; wants trends, high-risk segments and alerts when model quality drifts.
- **Integration Engineer:** Connects the scoring engine to the bank's loan-origination system; wants a stable, documented API.

➤ 5.2 Top User Journeys

- Loan officer: Login → Applicant queue → Open applicant → See risk band + top reasons → Approve / refer / decline. KPI: ≤ 4 clicks to a decision view; score visible ≤ 2 s.
- Risk manager: Login → Portfolio dashboard → Filter by product/period → Drill into high-risk segment → Export. KPI: dashboard load ≤ 2 s.
- Integration system: POST applicant JSON → receive score, band and reasons in ≤ 300 ms (p95)

➤ 5.3 User Stories

- As a loan officer, I want to see why an applicant is flagged so that I can explain my decision to the applicant and to auditors. GIVEN I open an applicant WHEN the score loads THEN I see the risk band and the top five contributing factors.
- As a risk manager, I want default-rate trends by month and segment so that I can tighten policy early. GIVEN I choose a date range WHEN I apply filters THEN charts update in ≤ 2 s.

➤ 5.4 Accessibility & Localization

- Keyboard-navigable core flows, AA color contrast, color-blind-safe risk colors (icons plus labels, not color alone); language: English (Hindi labels as a stretch goal).

6. Market and Competitors

➤ 6.1 Competitor Table

Competitor	Product type	Target users	Key features	Strengths	Weaknesses	Our differentiator
FICO Score	Credit scoring	Banks, lenders worldwide	Standardised scorecards	Industry standard, trusted	Proprietary, costly, less adaptive	Open, customisable ML with reason codes
Zest AI	ML underwriting	US lenders, credit unions	ML models, explainability	Mature, compliance-focused	Enterprise pricing, US-centric	Low-cost, India-oriented, open stack
Upstart	AI lending platform	Partner banks (US)	AI risk models, automation	Strong ML, scale	Closed platform, not India-focused	Pluggable API for any LOS
Scienaptic	AI credit decisioning	Banks, NBFCs (India)	Alt-data, decisioning	India presence	Enterprise-only, limited transparency	Transparent, student-built reference stack

➤ 6.2 Target Users / Customers and Market Segment

- **Primary customers:** Small and mid-sized banks, cooperative banks, NBFCs and microfinance institutions that need modern risk scoring but lack large in-house data-science teams.
- **Secondary customers:** Fintech and digital-lending apps, and credit teams inside larger banks piloting an ML challenger to their existing scorecard.
- **End users:** Loan officers, credit/risk managers, collections teams, and integration engineers.
- **Market segment:** Retail and personal loans first (largest volume, richest data), extendable to MSME, vehicle and education loans.

➤ 6.3 Market Research and Drives

- Asset quality is at a multi-decade best, but RBI stress tests show meaningful upside risk in adverse scenarios [1][2], which keeps early-warning and risk analytics on lenders' agenda.
- Growth of retail and digital lending increases application volume and the need for automated, consistent scoring.
- Regulators and auditors expect transparent, well-governed use of models, which favours explainable approaches over black-box ones.
- Existing solutions are mostly bureau scores (generic) or enterprise platforms (costly); small lenders are under-served.

➤ 6.4 Positioning

- **Unique angle:** An explainable, API-first, lender-specific default predictor with a usable officer dashboard, affordable for smaller institutions.
- **Measurable delta:** Target ≥ 10 points higher recall on defaulters than the logistic-regression baseline at the same approval rate; every score ships with reasons.

7. Objectives and Success Metrics

- **O1 Reduces Losses:** Back-test shows $\geq 10\text{--}15\%$ fewer defaults (or expected loss) at the same approval rate versus the baseline by 6 Dec 2026. KPI: default-rate reduction %.
- **O2 Predict risk early:** $\text{AUC-ROC} \geq 0.78$ and $\text{KS} \geq 0.35$ on held-out data; recall on defaulters $\geq 70\%$. KPI: AUC, KS, recall.
- **O3 Automate risk scoring / real-time:** p95 scoring latency ≤ 300 ms by 29 Nov 2026. KPI: p95 ms.
- **O4 Explainability:** 100% of scores include top-5 SHAP reasons. KPI: explanation coverage
- **O5 Integration:** Documented REST API (OpenAPI) with $\geq 95\%$ contract tests passing. KPI: test pass rate.

- **O6 Better lending decisions:** Officer dashboard usable in user testing with task-success $\geq 80\%$. KPI: SUS / task success.

8. Key Features

Features	Description	Priority	Dependencies	Acceptance Criteria
ML risk prediction model	Gradient-boosted model estimating probability of default	Must	Clean data, features	$AUC \geq 0.78$ on hold-out; calibrated probabilities
Data preprocessing	Validation, imputation, encoding, outlier handling, feature engineering	Must	Datasets	Pipeline reproducible; schema errors rejected with clear message
Real-time scoring	Low-latency scoring endpoint	Must	Model, API	$p95 \leq 300$ ms
Explainable predictions	SHAP reason codes for each score	Must	Model	Top-5 factors returned for every score
Loan officer dashboard	Applicant view, risk badge, explanation chart, queue	Must	API, Auth	Score visible ≤ 2 s; role-based access
Historical trend analysis	Default-rate and risk-mix trends by period/segment	Should	Stored Scores/Outcomes	Charts update ≤ 2 s on filter
API integration	Secured REST API for loan systems	Must	Auth, DB	OpenAPI docs; 201/200/400/401 codes verified
Monitoring & retraining	Drift and performance monitoring; retrain on new labels	Could	MLflow, Outcomes	Alert on $PSI > 0.2$ or AUC drop

9. Architecture

➤ 9.1 High-level

- **Clients:** React dashboard (loan officers, risk managers); external loan-origination system via API.
- **Services:** FastAPI gateway (auth, validation), scoring service, analytics service.
- **ML layer:** Preprocessing and feature pipeline, XGBoost/LightGBM model, SHAP explainer, MLflow registry and drift monitor.
- **Data stores:** PostgreSQL (applications, scores, outcomes), optional Redis cache, object storage (datasets and model artifacts).
- **Integrations:** Loan-origination system (REST/JSON); CSV batch import.

➤ 9.2 Technology Stack

Layer	Technology	Purpose
Data & ML	Python, pandas, NumPy, scikit-learn, imbalanced-learn	Preprocessing, baselines, evaluation
Models	Logistic Regression, Random Forest, XGBoost, LightGBM	Default probability prediction
Tuning & Tracking	Optuna, MLflow	Hyperparameter search, experiment/model registry
Explainability	SHAP	Global and per-applicant explanations
Backend	FastAPI, Pydantic, SQLAlchemy, Alembic	Scoring API, validation, ORM, migrations
Database	PostgreSQL (Redis optional)	Storage, caching
Frontend	React, Recharts (dark/light theme)	Officer dashboard and charts
DevOps	Docker, GitHub Actions, Render/AWS	CI/CD and hosting
Testing	pytest, pandera, Playwright, Locust	Unit, data, E2E and load tests

➤ 9.3 AI/ML Models and Methodology

1. Data: train on public credit datasets — Home Credit Default Risk (primary), Lending Club and UCI German Credit (validation/benchmark) [7][8][9]; synthetic data for demo records.
2. Preprocessing: schema validation, missing-value imputation, outlier capping, categorical encoding, scaling where needed.

3. Feature engineering: debt-to-income, loan-to-income, credit utilisation, payment-to-income, credit-history length, delinquency counts, Weight-of-Evidence/binning features.
4. Class imbalance: class weights / scale_pos_weight; SMOTE evaluated as an alternative [6].
5. Modelling: Logistic Regression (interpretable baseline), Random Forest, XGBoost [3] and LightGBM [4]; stratified 5-fold CV plus a time-based hold-out where dates exist; Optuna for tuning.
6. Calibration and thresholds: Platt/isotonic calibration; risk bands Low < 0.30, Medium 0.30–0.60, High > 0.60 (tuned on validation using a cost-sensitive threshold).
7. Evaluation: AUC-ROC, PR-AUC, KS statistic, recall/precision at threshold, Brier score, and a cost-based back-test of avoided losses.
8. Explainability and fairness: SHAP for reason codes [5]; sensitive attributes are excluded from features and approval/default rates are compared across groups as a bias check.
9. Monitoring: Population Stability Index (PSI) for drift, periodic re-scoring, retraining when outcome labels accumulate.

➤ 9.4 API Spec Snapshot

Endpoint	Method	Auth	Purpose	Request Schema	Response Schema	Codes
/api/v1/auth/login	POST	-	Obtain JWT	email, password	200 {token}	200, 401
/api/v1/score	POST	JWT/API Key	Score one applicant	applicant fields	200 {probability, band, reasons[]}	200, 400, 401
/api/v1/score/batch	POST	JWT/API Key	Score a CSV batch	file	202 {jobId}	202, 400
/api/v1/applicants/{id}	GET	JWT	Applicant + latest score	-	200 {applicant, score}	200, 404
/api/v1/trends	GET	JWT	Historical trends	from, to, segment	200 {series[]}	200, 400
/api/v1/decision	PATCH	JWT	Record officer decision	loanId, decision	200 {status}	200, 403

➤ 9.5 Config and Secrets

- .env for local development (Git-ignored); secrets in GitHub Actions/host secret store; model artifacts versioned in MLflow; rotate API keys each term

➤ 9.6 Scalability

- Stateless FastAPI containers scale horizontally behind a load balancer; the model is loaded in memory per worker for low-latency scoring.
- Redis caching for repeated lookups, database indexes on loan/applicant IDs, and read replicas for analytics queries.
- Batch scoring via a background queue (Celery/RQ) for large portfolios; feature computation is vectorised.
- Model registry supports A/B or champion–challenger rollout; multi-tenant configuration for multiple lenders as a later step.

10. Data Design

➤ 10.1 Data Dictionary

Entity	Field	Type	Null?	Allowed values	Source	Notes
Applicant	applicant_id	UUID	No	—	System	PK
Applicant	age	Integer	No	18–100	Applicant	Model feature
Applicant	annual_income	Decimal(12,2)	No	≥ 0	Applicant	Model feature
Applicant	employment_years	Decimal(4,1)	Yes	≥ 0	Applicant	Imputed if null
LoanApplication	loan_id	UUID	No	—	System	PK, FK applicant_id
LoanApplication	loan_amount	Decimal(12,2)	No	> 0	Applicant	Model feature
LoanApplication	tenure_months	Integer	No	6–360	Applicant	Model feature
RiskScore	default_probability	Float	No	0–1	Model	Calibrated
RiskScore	risk_band	Enum	No	Low/Medium/High	Model	From thresholds
Explanation	shap_value	Float	No	—	Model	Per feature, ranked
LoanOutcome	defaulted	Boolean	Yes	0/1	Bank/dataset	Label for retraining

➤ 10.2 Schema and Mitigations

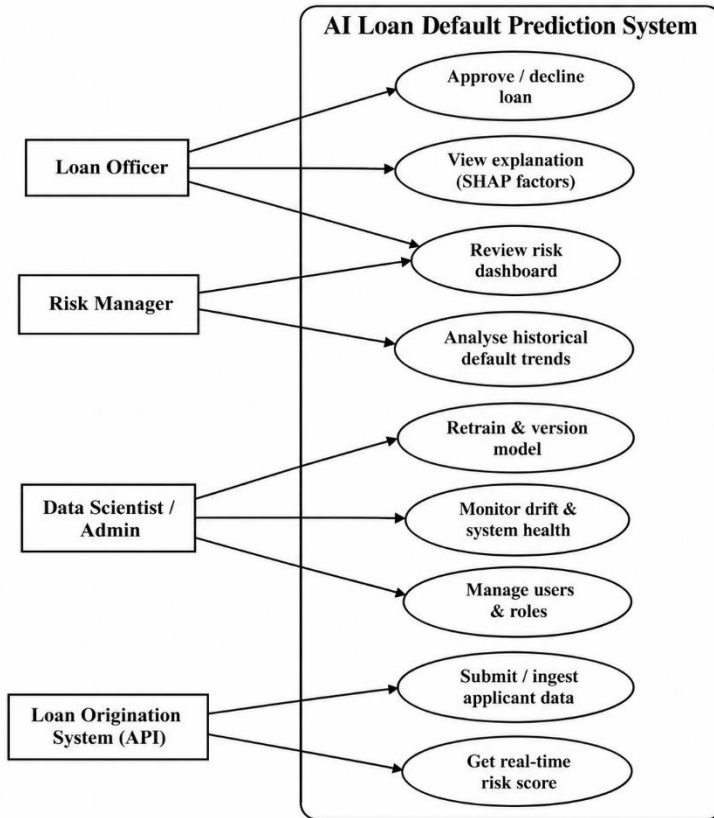
- ERD in Section 12; schema managed with Alembic migrations, versioned in Git; rollback tested on staging.

➤ 10.3 Privacy, Retention, Backup/DR

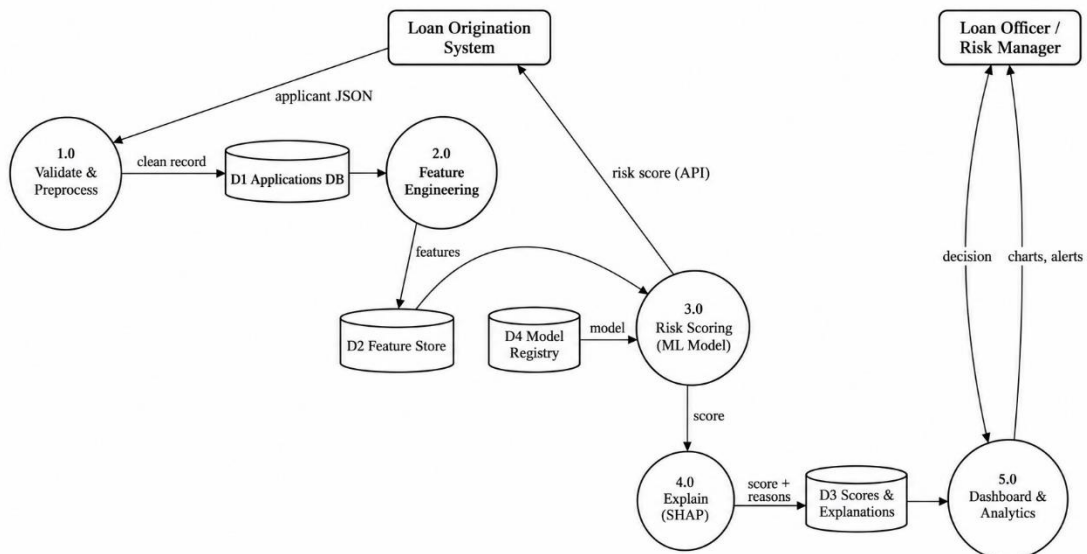
- **Only public/anonymised or synthetic data is used;** no real customer PII is stored. If PII (e.g., name, email of dashboard users) is stored, it is minimised and access-controlled.
- **Measurable Retention:** demo data refreshed each term; logs kept 90 days. Backups nightly; RTO: 4 h, RPO: 24 h.

11. Technical Workflow Diagram

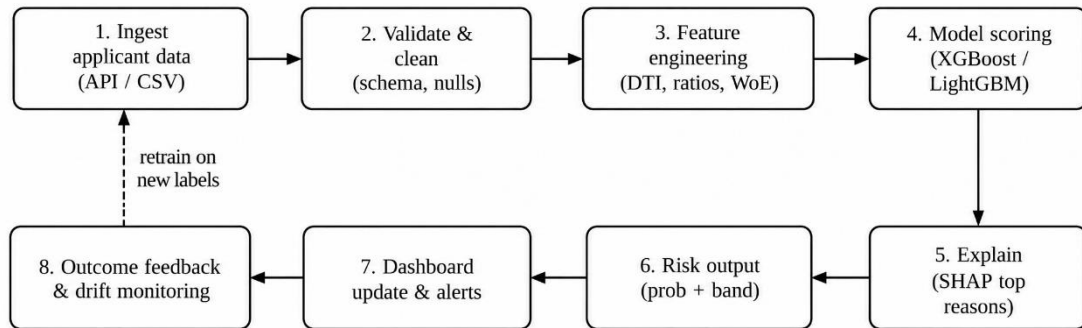
i. Use Case Diagram



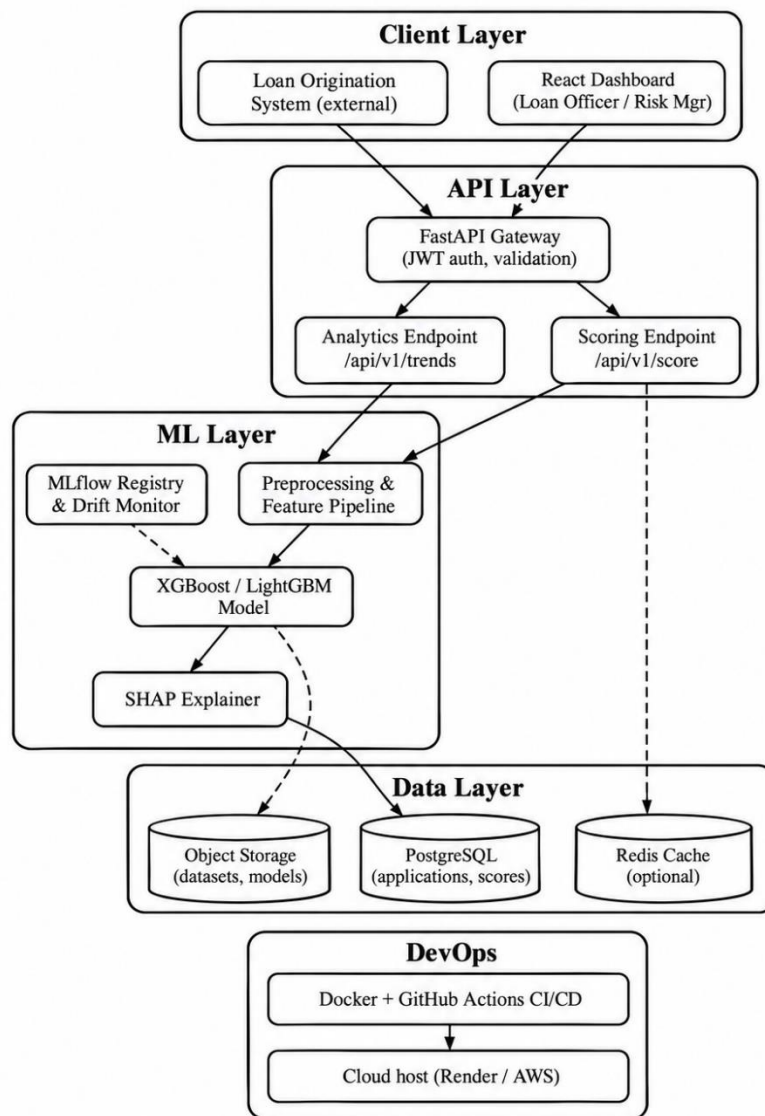
ii. Data Flow Diagram



iii. Technical Workflow Diagram



iv. System Architecture Diagram



12. Quality: NFRs and Testing

➤ 12.1 Non-functional Requirements

Metric	SLI	Target(SLO)	Measurement
Availability	Uptime %	$\geq 99.0\%$	Uptime monitor
Latency	p95 scoring latency	≤ 300 ms	APM / Locust
Dashboard	p95 page load	≤ 2 s	Lighthouse / APM
Error Rate	5xx %	$\leq 1\%$	Logs
Model Quality	AUC-ROC (hold-out)	≥ 0.78	MLflow report
Security	Open critical CVEs	0	Dependency scanner

➤ 12.2 Test Plan

Area	Type	Tools	Coverage Target	Exit Criteria
Data Pipeline	Unit + data validation	pytest, pandera	80%	No schema/leakage defects
Model	Validation & regression	scikit-learn metrics, MLflow	Metrics Gates	AUC/KS/recall targets met
Backend	Unit + integration	pytest, httpx	70%	No P1/P2 defects
UI	E2E + usability	Playwright, peer testing	60%	Pass rate $\geq 95\%$
API	Contract + load	Postman/Newman, Locust	All Endpoints	p95 ≤ 300 ms

➤ 12.3 Environments

- Dev → Staging → Prod; feature flags for risky changes; model promoted through registry stages (Staging → Production).

13. Security and Compliance

➤ 13.1 Threat Model (STRIDE)

Asset	Threat	STRIDE	Impact	Likelihood	Mitigation
Auth tokens	Token theft	Spoofing	High	Medium	HTTPS, short TTL, rotation
Applicant data	SQL injection	Tampering	High	Low	Parameterised queries, ORM, validation
Scores & decisions	Unauthorised change	Repudiation	Medium	Low	Audit log, role checks
Model	Model/data poisoning	Tampering	High	Low	Versioned data, validation gates, registry access control
API	Flooding / abuse	Denial of service	Medium	Medium	Rate limiting, request size limits

➤ 13.2 AuthN/AuthZ

- AuthN: email + password with JWT; API key for system-to-system calls.
- Roles: loan_officer, risk_manager, admin, service. Role checks at API layer.

➤ 13.3 Audit and Logging

- Log logins, score requests, officer decisions, model version used for each score; 90-day retention.

➤ 13.4 Compliance

- Academic project using public/synthetic data; follows institute policy. Design keeps data minimisation, explainability and auditability in mind for future alignment with RBI guidance and India's DPDP Act, 2023 [21][22].

14. Delivery and Operations

➤ 14.1 Release Plan

- v1.0 demo in the week of 7 Dec 2026; internal alpha (weeks 5–6), integrated beta (week 8), release candidate (week 9).

➤ 14.2 CI/CD and Rollback

- CI: lint → unit tests → data/model checks → build → dockerise; CD to staging on main; rollback via previous image tag and previous model version.

➤ 14.3 Monitoring and Alerting

Metric	Threshold	Alert to	Runbook
p95 Scoring latency	> 500 ms	Backend	“API Latency”
Error Rate	> 2%	Backend	“Error Spike”
Data Drift	>0.2%	ML	“Model Drift”
Live AUC / Default Capture	Drop > 5%	ML	“Model Drift”

➤ 14.4 Runbooks

- **API Latency:** check DB indexes and cache hit rate → scale containers → roll back last change.
- **Error Spike:** inspect logs → roll back image → write incident note.
- **Model Drift:** confirm data shift → retrain on recent labels → validate → promote via registry.

➤ 14.5 Communication Plan

- Stand-ups Mon/Wed/Fri; weekly status to mentor each Friday; bi-weekly demo

➤ 14.6 Deployment Plan

- Containerise API and model with Docker; deploy frontend to Vercel/Netlify and API to Render or AWS (ECS/EC2); managed PostgreSQL (Neon/RDS).
- Promote Dev → Staging → Prod through GitHub Actions; keep model artifacts and versions in MLflow; secrets in the host's secret store.
- Real-world rollout path: shadow mode alongside the existing scorecard → champion–challenger → decision support for officers.

15. Risks and Mitigations

Risk	Probability	Impact	Score	Mitigation	Status
Class imbalance / weak recall	High	High	16	Class weights, threshold tuning, PR-AUC focus	Open
Data leakage inflating metrics	Medium	High	12	Time-based split, leakage checks, peer review of features	Open
Schedule slip	Medium	High	12	Scope freeze, weekly demos	Open
Public data not representative	Medium	Medium	9	State limits, use multiple datasets	Open
Latency above target	Low	Medium	6	Model size limits, caching, profiling	Open
Bias / fairness concerns	Medium	High	12	Exclude sensitive features, group metrics report	Open

16. Research and Evaluation

- Market review: FICO, TransUnion CIBIL, Zest AI, Upstart and Scienaptic studied for approach, explainability and target customers [15]–[19].
- Evaluation plan: track KPIs in Section 8 weekly; compare each model to the logistic-regression baseline; run a cost-based back-test showing defaults avoided at equal approval rate; user testing with 3–5 peers/faculty acting as loan officers.
- Limitations: public datasets rather than real bank data; results are indicative; no live bureau integration; fairness analysis limited to available attributes.

➤ 14.6 Future Scope

- Integrate credit-bureau and Account Aggregator data for richer features.
- Behavioural/sequential models (e.g., temporal models on repayment history) for early-warning of existing loans.
- LLM-generated plain-language adverse-action and explanation summaries.
- Fairness and bias dashboard; formal model-risk documentation.
- Probability-of-default output feeding expected-credit-loss (ECL) estimation.
- Multi-tenant SaaS deployment, mobile companion app, and Hindi/regional-language UI.
- Federated learning so lenders can improve models without sharing raw data.

17. Appendices

- **Glossary:** PD (probability of default), NPA/GNPA, AUC-ROC, KS (Kolmogorov–Smirnov), PSI (Population Stability Index), SHAP, DTI, WoE, LOS (loan-origination system), JWT, SLO, p95.

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